



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

**FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS

MMA 109: FOUNDATION OF PURE MATHEMATICS

Date: 28th January, 2025

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- DO NOT write anywhere on this question paper
- Answer Question ONE and any other TWO.
- Start each question on a fresh page
- Observe further instructions on the answer booklet



MMA 109: FOUNDATION OF PURE MATHEMATICS

90698

[30 Marks]

QUESTION ONE (Compulsory)

- (a) In mathematics, how to write is as important as what to write. Here's a student's proof of the following theorem:

Theorem 1. *If n^2 is an even integer then n is an even integer.*

Proof.

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2m + 1.$$

Criticize the student's proof and rewrite it.

[5 Marks]

- (b) Write the converse, inverse and contrapositive of the sentence:

If x is an even integer greater than 2 then it is not a prime number.

[3 Marks]

- (c) Prove the following statements:

(i) 6 is not an odd integer.

[2 Marks]

(ii) If pigs can fly then $5^2 = 25$.

[2 Marks]

(iii) If 6 is a prime number then $7^2 = 50$.

[2 Marks]

(iv) If x and y are integers then $7x + 21y \neq 2024$.

[2 Marks]

(v) $\sqrt{2}$ is an irrational number.

[4 Marks]

- (e) Negate the statement: If Maseno University lies on the equator then Siriba Campus is not in the Southern hemisphere and Maseno University is warm.

[2 Marks]

- (f) Explain the difference between these two statements, identifying which one is true, if we assume the universe to be $\mathbb{R}$.

First Statement: $\forall x, \exists y$ such that $x^2 - y = 0$.

Second Statement: $\exists y$ such that $\forall x, x^2 - y = 0$.

[3 Marks]

- (g) Define a relation on the real numbers $\mathbb{R}$ by $x \sim y$ if and only if $x - y$ is divisible by 2. Show that this relation is an equivalence relation.

[5 Marks]

QUESTION TWO

[20 Marks]

- (a) Use the principle of mathematical induction to prove that a regular polygon with n sides for $n \geq 3$ has a total internal angle sum of $180(n-2)$ degrees.

[4 Marks]

- (b) Prove that $\{x \in \mathbb{R} : x^2 - x - 6 = 0\} = \{-2, 3\}$.

[4 Marks]

- (c) Use proof of negation to show that $2^{333} + 1$ is not prime.

[4 Marks]

- (d) Suppose that x and y are real numbers. Prove that if $(x+1)(y^2+1)$ is odd then x is even or y is odd.

[3 Marks]

- (e) Give a 'picture proof' to the inequality: $a^2 + b^2 \geq 2ab$.

[3 Marks]

QUESTION THREE

[20 Marks]

- (a) Differentiate between *tautology* and *contradiction* and hence use a truth table to show that the statement $(A \Rightarrow \neg B) \Leftrightarrow (\neg A \vee \neg B)$ is a tautology. [5 Mark]
- (b) By the use of quantifiers, connectives and logically equivalent statements, negate the statement "People who live in glass houses or do not live in the air should not throw stones and dance with the angles." [6 Mark]
- (c) Use telescoping to show that: $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{99} + \sqrt{100}} = 9$. [5 Mark]
- (d) Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = 2x + 5$. Show that f is a bijection. [4 Mark]

QUESTION FOUR

[20 Marks]

- (a) Define an *axiom* and hence give any three axioms of geometry as given by Euclid. [4 Mark]
- (b) Let A and B be sets. Give an axiomatic proof of the statement: $A - (A \cap B) = A - B$. [3 Mark]
- (c) Define a *paradox* and hence explain potato paradox. [4 Mark]
- (d) Use contradiction to prove that the sum of a rational number and an irrational number is an irrational number. [3 Mark]
- (e) Let x be an integer. Prove that x^2 divides x if and only if $x \in \{-1, 0, 1\}$. [6 Mark]

QUESTION FIVE

[20 Marks]

- (a) A student conjectured that $1=2$ and attempted to prove it. Here's the proof.

Proof. We start with the obvious equality

$$a^2 - a^2 = a^2 - a^2 \quad (1)$$

Since $x^2 - y^2 = (x + y)(x - y)$, then we use this fact to factorize the right hand side of (1) to obtain

$$a^2 - a^2 = (a + a)(a - a) = 2a(a - a).$$

Moreover, we factor out a on the left hand side to get $a(a - a) = 2a(a - a)$. Dividing both sides by $a(a - a)$, we obtain $1 = 2$. $\square$

Identify the mistake(s) in the proof.

[2 Marks]

- (b) Explain George Pólya's four-step process of problem solving. [4 Marks]
- (c) Prove that for all natural numbers n , the expression $n^3 + 3n^2 + 2n$ is a multiple of 6. [5 Marks]
- (d) Prove in cases that if $a, b \in \mathbb{Z}$ and $a = b^3$ then there is an integer c such that $a = 9c$, $a = 9c - 1$ or $a = 9c + 1$. [9 Marks]